

ACTIONABLE PLAN

Week 2 Task List (27 July – 3 August 2026)

1. Scope & Baseline Assumptions

- ☐ **Confirm Zero-Grid Baseline Assumption:** State clearly in project documentation that any standalone solar panels or power lines visible in site satellite imagery are excluded from baseline calculations to maintain a strict zero-grid baseline per course brief rules.
- ☐ **Align Structure Terminology with Official Census:** Consult official national census classifications to accurately categorize dwelling types (e.g., traditional mud-walled structures, thatched units, corrugated roofs) rather than using generic terms like "household".

2. Load Profiles & Energy Demand

- ☐ **Develop Literature-Backed Load Profiles:** Establish daily energy demand, appliance power ratings (W), and time-of-use schedules aligned with target ESMAP access tiers and supported by energy access literature—grounding appliance adoption in realistic rural ownership patterns rather than assuming standard AC consumption.

3. Competitor Failure Analysis & Physical Security

- ☐ **Document Competitor Failure Modes:** Evaluate existing off-grid energy providers and document specifically where their models struggled (e.g., installation cost overruns, theft, maintenance difficulties) to justify how the proposed design addresses these issues.
- ☐ **Design Analogue Physical Security:** Develop hardwired, analogue anti-theft mechanisms (such as internal pole-cut sensing wire loops or magnetic tamper switches) that operate reliably without cellular or internet connectivity.

4. Stakeholder & Community Strategy

- ☐ **Identify Local Leadership Structures:** Map out local administrative and traditional leadership hierarchies to establish a clear framework for community engagement and local support.

5. Deliverables & Engineering Notebooks

- ☐ **Log Progress in Engineering Notebooks:** Record all Week 2 findings, literature citations, load calculations, and supervisor feedback in individual digital engineering notebooks.